Before the Warning Labels: The Obscured Ethics of Generative AI Adoption

Suhayb Faran Mir

March 31st, 2026

Abstract

The rapid adoption of generative artificial intelligence has been accompanied by narratives of productivity, creativity, and innovation, while the broader ethical costs of these systems remain largely unexplored. This paper examines the hidden consequences of generative AI across multiple dimensions, including unreliable sourcing, environmental degradation, artistic exploitation, privacy erosion, guardrail failure, and the erosion of learning through cognitive offloading. Rather than treating these issues as isolated failures, the analysis identifies a recurring structural pattern in which benefits are immediate and widely distributed, while harms are cumulative, externalised, and disproportionately borne by individuals and communities with limited power of choice.

Drawing on recent reporting, technical analysis, and real-world case studies, the paper demonstrates how generative AI systems can produce fluent but poorly grounded outputs, enable harmful misuse when safeguards fail, and impose environmental and social costs through large scale data centre infrastructure. Particular attention is given to the illusion of safety and privacy created by conversational interfaces, as well as the implications of widespread reliance on AI tools in educational and professional contexts.

The paper does not argue for the abandonment of generative AI, nor does it seek to prescribe corporate or regulatory solutions. Instead, it adopts an individual-centred perspective, emphasising informed use, moderation, and critical engagement. By making the hidden mechanisms and consequences of generative AI visible, the paper aims to shift its use from default behaviour to conscious choice. As with earlier technologies whose harms only became apparent after widespread adoption, ethical reflection is most urgent before formal warning labels arrive.

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1. Introduction

The year 2022 marked a significant technological milestone with the public release of OpenAI's generative AI model, ChatGPT. Within months, usage of generative AI tools increased exponentially, reaching millions of users globally (Hu, 2023). Today, only a few years later, AI has become embedded in daily life: individuals use it for tasks as simple as a search query, and even Google now integrates generative AI directly into its core search algorithms (Google, 2024). It is increasingly clear that AI is here to stay, even though the world may suffer unintended consequences if its adoption is left unchecked and unregulated.

To shift perspective, history provides a cautionary tale. In the 1920s, cigarette companies marketed smoking as fashionable and even health-enhancing, while suppressing evidence of long-term harm (Proctor, 2012). It was only after undeniable correlations emerged between smoking and cancer that regulators required health warnings on packaging. Generative AI, much like tobacco a century ago, is marketed as an innovation that empowers, entertains, and increases productivity, while its hidden costs, from environmental damage to ethical risks, remain obscured.

A central ethical problem underlying this rapid adoption can be described as the myth of consent. In a technical sense, users do consent to generative AI systems through terms of service, privacy policies, and default platform settings. However, this consent is rarely informed in any meaningful way, and these policies never depict the full picture. Users typically engage with these tools before understanding how their data may be used, what environmental or social costs are incurred, or how deeply these systems are integrated into broader corporate, governmental, and military infrastructures. By the time these realities become visible, opting out is often impractical or impossible, as the data has already been processed, models have already been trained, and the systems themselves have become embedded into everyday workflows.

As a student when ChatGPT first entered the public domain, I experienced first-hand benefits as well as its destructive potential in educational contexts. On the initial release, ChatGPT felt like a groundbreaking tool. I remember my first query was asking it to code the classic Fibonacci sequence problem and it provided the correct results in less than 2 minutes. I felt like I was ready to take on the world with this tool by my side. However, I quickly realised – if I don't use this tool in moderation, I will lose many of my skills that have taken me this far into the institution in the first place.

During my final years at university, I completed a dissertation on the citation accuracy of AI chatbots (ChatGPT 3.0 and Claude 2.0) in an academic setting with particular focus on electrical engineering. The main takeaway from the paper was that these AI tools can provide correct sources only 50 percent of the times. Things have changed since then and now some models, with the help of external tools, are able to view data in real time over the web. However, the citation accuracy is still a major issue, as confirmed when Deloitte was forced to apologise for an Australian government report that contained fabricated citations and AI generated errors. The firm admitted that "oversight was not followed" and reissued the report after fake academic references and a made-up Federal Court quote were discovered (Tadros, 2025). This incident highlights how even leading professional organisations can fail to verify AI outputs properly, reinforcing the ethical need for transparency, human oversight, and accountability in all AI assisted work.

This paper aims to examine six key ethical dimensions of generative AI: sourcing capabilities, environmental and corporate impact, artistic theft, privacy concerns and guardrail failure, and learning restrictions.

AI will remain a permanent fixture in human society. The pressing challenge is ensuring it is adopted responsibly, with awareness of its hidden consequences. By the end of this paper, the reader should be equipped to critically reflect on their own usage of generative AI. Before performing even a simple query, ask yourself: "Do I really need AI for this particular task?"

1.1 What is Generative AI?

It is vital that before delving into the ethical considerations, we must define generative AI. In this paper, every use of the term AI will refer to generative AI, unless stated otherwise. Everyone has their own definition of generative AI, but to remain consistent, Stefan Feuerriegel explains it as "Computational techniques that are capable of generating seemingly new, meaningful content such as text, images, or audio from training data (Feuerriegel, 2023)." Essentially, Gen AI is a tool that can create new forms of text, speech, art, etc. using previously trained data.

1.2 Differences between Generative AI and regular AI

The distinction between generative AI and regular AI lies primarily in its purpose and output. Traditional, or "narrow," AI systems are designed to analyse data and make predictions or classifications based on patterns they have learned. For instance, some usage of traditional AI could be recommending movies, detecting fraud, or recognising faces all based on various algorithms. Generative AI, on the other hand, goes a step further by creating new data that resembles its training input. According to the University of Illinois, "Generative AI models often require substantial computational resources and training time, making it more difficult to scale compared to traditional AI, which is typically more efficient—particularly when performing well-defined tasks that require less extensive model training (OoC, 2024)."

1.3 Positionality and Methodological Approach

This paper adopts a qualitative, synthesis-based approach to examining the ethical implications of generative artificial intelligence. Rather than conducting original empirical experiments, the analysis draws on a combination of technical documentation, investigative journalism, academic literature, policy reports, and real-world case studies. These sources are analysed collectively to identify recurring structural patterns in how generative AI systems are developed, deployed, and experienced.

The use of first-person perspective in this paper is intentional. My engagement with generative AI spans academic research, professional use, and direct evaluation of AI systems in applied contexts. This positionality allows the paper to connect abstract ethical principles with lived experience and documented outcomes, particularly in domains where harms are diffuse, indirect, or externalised.

The paper does not claim causal certainty in complex socio-technical systems, nor does it attempt to quantify all impacts precisely. Instead, it aims to surface ethical risks that emerge repeatedly across different contexts, even when formal accountability or attribution remains contested. In this sense, the contribution of the paper is interpretive and normative: it seeks to make visible ethical consequences that are often obscured by narratives of innovation, convenience, and inevitability.

2. Scope

To remain consistent, this paper focuses specifically on the ethical considerations of generative AI rather than artificial intelligence itself. The analysis does not examine every branch of machine learning, nor does it attempt to cover the full range of predictive, surveillance, or statistical AI systems used in industry and defence. Within generative AI, the discussion concentrates on large-scale, widely deployed systems that currently dominate public, enterprise, and governmental use, rather than attempting an exhaustive survey of all existing or experimental models.

However, certain non-generative AI examples are referenced briefly when they help illustrate broader ethical patterns that may emerge as generative AI becomes more powerful and more deeply embedded in society. For instance, companies such as Palantir do not build generative models, yet their involvement in military operations demonstrates how advanced AI technologies can become tightly intertwined with state power and conflict. These cases are included not to expand the scope into general AI, but to highlight the potential pathways and risks that generative AI companies may follow as their capabilities grow.

The central aim of this paper is therefore to examine the ethical risks created by generative AI systems and their developers, with particular attention to those models that shape real-world behaviour at scale, while using selected comparative examples only when they clarify how these risks may intensify or evolve in current world contexts.

3. Architecture of Generative AI

What Happens When You Submit a Single Query?

Before examining sourcing failures, environmental cost, or ethical risk, it is necessary to understand at a high level what a single user query triggers inside a generative AI system. Although the interface looks like a simple chat box, each prompt initiates a pipeline that (1) converts text into tokens, (2) runs a trained neural network to generate output token-by-token, (3) applies safety controls, and (4) returns a response to the user. Understanding this pipeline matters because the same architectural features that make LLMs powerful also explain why they can produce fluent inaccuracies, why safety controls are imperfect, and why privacy and data retention are central ethical concerns (Bender et al., 2021).

General Architecture of a Single Query		
Stages	What Happens	Sources

User Input & Transmission	A user submits a text prompt through an interface. The prompt is transmitted to the AI provider's servers (data centre) for processing as part of a request to a hosted language model.	OpenAI Platform (a) & Zewe (2025)
Tokenisation	The text prompt is split into tokens (sub-words or symbols). Tokens are the basic units processed by the model. The final response is also generated and returned as tokens before being converted back into text.	OpenAI (2025e)
Embeddings & Model Architecture	Tokens are mapped into numerical embeddings and processed by a Transformer-based neural network that models relationships across the full input sequence.	Vaswani et al. (2017)
Inference (Response Generation)	The model generates output autoregressively, predicting the most likely next token at each step based on learned probability distributions until a full response is produced.	Jurafsky & Martin (2025)
Safety & Moderation Layers	Generated outputs are evaluated by moderation systems and policy enforcement mechanisms before being returned to the user. These systems operate alongside the core model.	OpenAI Platform (b)
Post-Response Handling (Retention & Training)	Depending on the product and settings, prompts and responses may be logged temporarily for safety monitoring. API inputs are not used for training by default; consumer chat data may be unless users opt out.	OpenAI (2026b)

Table 1- High-level lifecycle of a single generative AI query

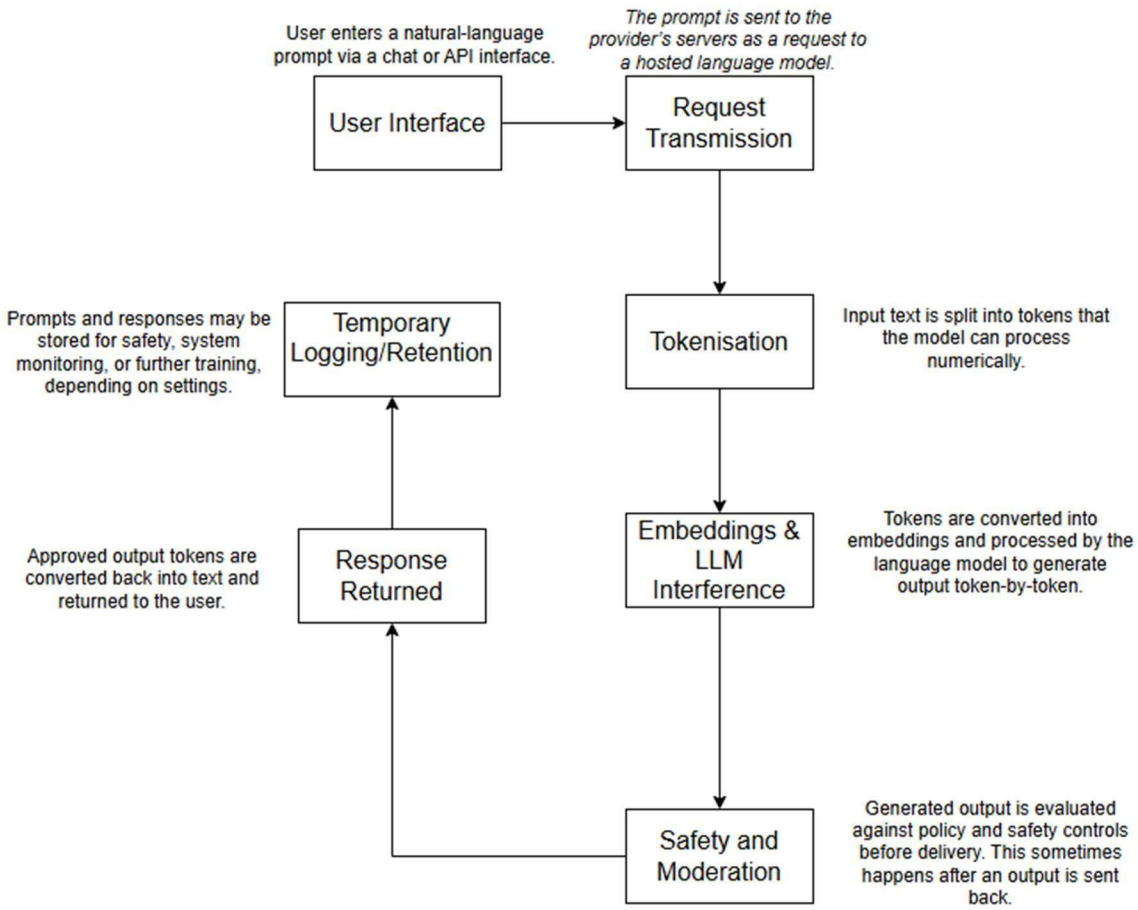


Figure 1 - Architecture Process Flow

Table 1 and Figure 1 together provide a concise, high-level view of how a single generative AI query is processed from initial user input to final response delivery. The table outlines each stage of the pipeline in a structured manner, while the accompanying figure visualises the same sequence as a continuous flow, highlighting the order in which tokenisation, model inference, moderation, and response handling occur. This architectural grounding establishes a shared technical baseline for the ethical analysis that follows, ensuring that later discussions are anchored in an accurate understanding of how generative AI systems operate in practice.

4. Sourcing Capabilities

In 2023, I began writing my honours thesis titled "Ethics of AI Chatbot's Sourcing Capabilities." The central argument explored how LLMs such as ChatGPT and Claude 2 obtain and represent knowledge. Through testing and analysis, I found that these models, while seemingly authoritative, often struggle with the origins and accuracy of their information. Although this paper was written back in 2023 when these tools were new and have since been replaced by newer models, the main takeaways still stand to this day.

4.1 Data Origins

Generative AI models like ChatGPT and Claude are not explicitly trained on databases paired with their original sources. Instead, they learn from vast collections of text gathered from the internet, third-party partnerships, and human contributions. OpenAI confirms this in its documentation: "OpenAI's foundation models, including the models that power ChatGPT, are developed using three primary sources of information: (1) information that is publicly available on the internet, (2) information that we partner with third parties to access, and (3) information that our users, human trainers, and researchers provide or generate (OpenAI, 2025a)."

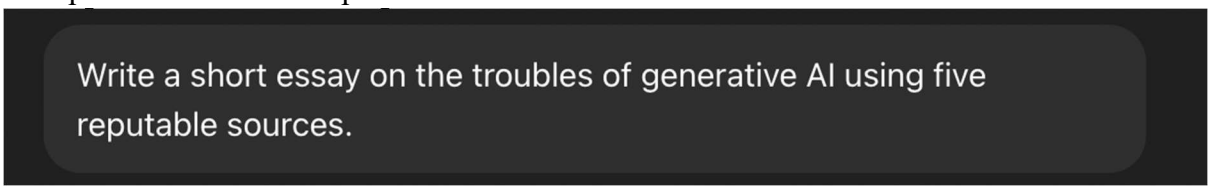
This statement demonstrates that LLMs do not inherently retain or cite the original sources from which the information was learned. During my thesis research, I tested this limitation by prompting ChatGPT with academic-style queries and evaluating the validity of its references. The results showed that only about 50 percent of the time the model produced a legitimate and verifiable citation; the other half of the time, it generated incorrect, outdated, or completely fabricated references.

4.2 Fact-Checking After Retrieval

Since the training data of these models lacks explicit source-to-content mapping, the transparency of information retrieval becomes a major ethical concern. When users ask for factual content, the LLM retrieves patterns that resemble the requested information rather than fetching it from traceable documents.

This makes post-retrieval verification difficult. If an answer is incorrect, there is no clear trail to determine where the information originated or whether it was influenced by unreliable data. In my tests, this issue appeared most prominently when prompts did not instruct the model to access the web.

Example 1: Incorrect Prompt



In this prompt, the LLM is not explicitly instructed to acquire information by browsing the web. Instead, the prompt encourages the model to generate a response using its internal parameters, which are derived from patterns learned during training rather than from real-time information retrieval. Importantly, this does not imply that the model "thinks for itself" in a human sense. Large language models do not engage in reasoning or independent thought; they operate as probabilistic token-generation systems that predict likely sequences of words based on prior training.

Additionally, while LLMs can reference concepts, styles, and factual patterns learned during training, they are not trained on, nor do they have access to, specific source documents or databases. As such, any information produced is a reconstruction based on learned statistical relationships, not direct recall or consultation of original sources.

Example 1: Incorrect Prompt's Results for References

References

- Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). *On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?* Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency.
- Vincent, J. (2023). *The Hidden Costs of Generative AI and Deepfakes. The Verge.*
- Kastrenakes, J. (2023). *Artists Sue AI Companies Over Copyright Violations. The Verge.*
- Strubell, E., Ganesh, A., & McCallum, A. (2019). *Energy and Policy Considerations for Deep Learning in NLP. Proceedings of ACL 2019.*
- Floridi, L., & Chiriatti, M. (2020). *GPT-3: Its Nature, Scope, Limits, and Consequences. Minds and Machines*, 30(4), 681–694.

When an LLM is instructed not to browse the web, it does not retrieve information from stored sources or databases. Instead, it generates responses by predicting text based on statistical patterns learned during training. As a result, any citations it produces are not retrieved from memory or verified repositories but are constructed outputs that may resemble real references. This explains why, in the results above, only three of the five cited references correspond to real sources, while the remaining two are fabricated. This outcome supports the thesis finding that, without explicit constraints or verification mechanisms, LLMs are unreliable at producing accurate citations. To improve source validity, prompts must be carefully structured to either restrict citation behaviour or require externally verifiable references.

4.3 Hallucinations

One of the major issues with generative AI is hallucination, where models produce outputs that are fluent and confident but factually incorrect or entirely fabricated. Because these systems are trained on vast datasets without the original sources being recorded, the model often "fills in the gaps" by predicting likely-sounding text rather than retrieving verifiable facts. This is why you'll sometimes see random answers or gibberish citations like I observed in my tests. For example, research shows that large language models frequently generate content that is fluent and plausible-looking but lacks grounding in reliable evidence or external verification (Alansari & Luqman, 2025).

In practical terms, the risk of hallucination becomes serious when AI is used in high-stakes or professional contexts. The Deloitte case in Australia is a stark illustration. Deloitte Australia was contracted by the Department of Employment and Workplace Relations (DEWR) to produce a compliance review and assurance report (approximately AU$440 000) on the department's welfare and payment systems. The 237-page report was published in July 2025 and billed as a full audit of compliance and IT systems. However, serious issues were later discovered in the document, including fabricated academic citations, references to non-existent journals and authors, and even a misattributed quote from a federal court judgment (McGuirk, 2025).

Deloitte subsequently admitted that portions of the work had been assisted by a generative AI system (specifically an Azure OpenAI model) and agreed to refund the final instalment of the contract. The department republished a corrected version of the report, stripping out more than a dozen bogus references and rewriting sections, but stated that the "substance" of the report's findings remained unchanged (McGuirk, 2025). The incident demonstrates how hallucinations in LLM's outputs can creep into official documentation and undermine the integrity of professional and governmental decision-making.

When large language models are used without strong human oversight, verification, and source tracing, they can introduce errors that are not merely semantic but structural. For example, the prompt below does not explicitly state that the LLM should acquire information by using tools to surf the web. Using this prompt allows the model to think for itself and use sources in its database, which we have learned is not trained with the training data.

Example 1: Incorrect Prompt's Results for References

References

- Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). *On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?* Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency.
- Vincent, J. (2023). *The Hidden Costs of Generative AI and Deepfakes. The Verge.*
- Kastrenakes, J. (2023). *Artists Sue AI Companies Over Copyright Violations. The Verge.*
- Strubell, E., Ganesh, A., & McCallum, A. (2019). *Energy and Policy Considerations for Deep Learning in NLP. Proceedings of ACL 2019.*
- Floridi, L., & Chiriatti, M. (2020). *GPT-3: Its Nature, Scope, Limits, and Consequences. Minds and Machines*, 30(4), 681–694.

When an LLM model is told not to browse the web for information, it retrieves information from sources that it has already stored in memory. From the results above, only three of the references from the five are valid sources while the other two just do not exist – proving my thesis results that these LLMs can only cite the correct source of information about half of times. To correctly obtain the information with valid sources, the prompt must be phrased differently. As noted before, because the underlying AI training does not preserve source-to-content mappings, users cannot reliably trace where the information originated or whether it was plausible to begin with. That means hallucinations are not just minor glitches, but they strike at the heart of trust in AI generated knowledge.

The Deloitte episode reinforces the ethical need for transparency, human verification, proper prompting and accountability in all AI-assisted work. It also shows that even leading professional organisations can fail to implement adequate controls when using generative AI. Users and institutions must not assume that fluency equals factuality.

4.4 Verification Strategies

Modern LLMs, such as OpenAI's ChatGPT with browsing enabled and Anthropic's Claude 3, can now surf the web to gather information in real time. However, the reliability of the output still depends heavily on how the user formulates the prompt.

Example 2: Correct Prompt

Using five sources as references, write a short essay on the dangers of generative AI

In this prompt, the LLM is being asked to use the references as the direct source of information instead of having to pull out information and match that with a source. This way, the LLM knows that it must browse the web and obtain correct sources.

Example 2: Correct Prompt Results for References

References

1. "Safety and security risks of generative artificial intelligence to 2025 (Annex B)", UK Government. GOV.UK
2. Alan T. Yang & Andrew T. Yang, *Social Dangers of Generative Artificial Intelligence: Review and Guidelines*, ACM DGO 2024. ACM Digital Library
3. "Six Risks Of Generative AI" – Forbes. Forbes
4. Glen Weyl, "The Real Dangers of Generative AI", *Journal of Democracy*. Journal of Democ...
5. "Risks and Unknowns of Generative AI", OECD.AI. oecd.ai

The LLM accurately searches the internet to obtain five sources, validates that the sources are about the relevant topic, and then writes an essay based on the references. All five of the sources provided in the answer are legitimate and the information in the essay can be derived from them. Prompting is a skill itself, as it can teach users how to get optimal results.

These skills to prompt are essential, especially in a learning sense. If people use these models, but can't verify the sources, how can they claim the information given to them is accurate? This was the primary reason as to why I chose to write my honours thesis on the sourcing capabilities of AI chat bots.

5. Environmental and Large Company Impact

This section arguably highlights the most ethically concerning dimensions of generative AI. When people use these LLM tools today, there appears to be limited understanding of the impact of their actions and what these AI companies represent. No matter how good a product is, if a company's reputation gets tarnished by controversies, the demand will fall.

Looking at an analogy in a non-technical domain, Nestlé is a company that can be directly compared to big-tech. For years Nestlé was seen as a household name trusted by families all over the world. Its marketing portrayed the company as caring about health, nutrition and community. However, behind the scenes, the company was involved in serious ethical

controversies, including child labour in cocoa farms (Boffey, 2015) and water extraction during droughts (Milman & Luscombe, 2019).

Nestlé's situation demonstrates how a global corporation can enjoy decades of consumer trust while serious ethical issues remain hidden. Once investigations exposed child labour concerns, water rights issues, and supply chain abuses, consumers were confronted with a clear choice: continue purchasing Nestlé products or walk away. Boycotting Nestlé is, in practical terms, straightforward as people can easily switch to alternatives that are produced more ethically.

In contrast, companies such as OpenAI and Anthropic operate in a space where meaningful avoidance is becoming harder. Their models and APIs are embedded into corporate, government, and consumer tools. For example, Amazon Web Services (AWS) invested up to US $4 billion in Anthropic, making AWS its primary cloud and model-training partner (Anthropic, 2024). Likewise, OpenAI has formed enterprise and public-sector partnerships with companies such as SAP. SAP's enterprise AI co-pilot, Joule, is powered by third-party foundation models including OpenAI, PaLM 2, Gemini, Anthropic, Llama, and Mistral AI, alongside SAP-specific models like the ABAP LLM (SAP, 2024). Additionally, SAP and OpenAI have launched a sovereign "OpenAI for Germany" service through SAP's Delos Cloud and Microsoft Azure infrastructure (SAP, 2025).

This shows how a small group of foundation-model providers now dominate the enterprise AI ecosystem, creating a situation where their influence is centralised and their technology becomes deeply integrated across industries. With Nestlé, consumers retained the power to disengage. With generative-AI providers, the power imbalance is far greater and the transparency and oversight much weaker. This is worrying as generative AI will only continue to grow from now on.

In short: if you could avoid buying Nestlé's chocolate, you could make a choice; but you cannot realistically avoid using or being affected by the AI systems now embedded in search engines, enterprise software, government tools, and consumer apps. Try doing a Google search today without having an AI overview being shown first. Even YouTube videos have AI generated summaries that are depicted before a video even starts. This intensifies the ethical need for awareness, regulation, transparency, and user autonomy. The sections that follow begin by examining the environmental, infrastructural, and corporate impacts of this technological monopoly and dominance.

5.1 Data Centres and Electricity Usage

To understand the environmental scale of generative AI, we must first understand what an AI data centre is. A data centre is a temperature-controlled facility that contains vast amounts of computing infrastructure such as GPUs, servers, storage systems and networking equipment. These buildings are not small or simple. Amazon alone operates more than one hundred data centres globally, each holding roughly fifty thousand servers dedicated to cloud computing operations (Zewe, 2025). While the world has used data centres since the 1940s, the rise of generative AI has completely changed their scale and intensity.

The primary difference today is the sheer power density required for AI training and inference. As Noman Bashir from MIT's Climate and Sustainability Consortium explains, a generative-AI training cluster can consume seven or eight times more energy than a typical computing workload. This is because hyperscale AI centres, operated by large cloud service providers known as hyperscalers (such as Google, Microsoft, and Amazon), are facilities designed to run tens of thousands of servers at massive scale and high utilisation. These centres are filled with tightly packed racks of GPUs, CPUs, and specialised accelerators operating around the clock to train and continuously update large models (Zewe, 2025).

This acceleration in demand is already visible in global electricity figures. Data centres collectively consumed about 415 terawatt-hours of electricity in 2024, which accounts for roughly 1.5 percent of the world's total electricity usage. The International Energy Agency projects that this figure could more than double by 2030, reaching approximately 945 terawatt-hours, with AI as one of the dominant drivers of this growth (IEA, 2025).

Certain countries already illustrate what this future looks like. In the United States, data centres used around 176 terawatt-hours in 2023, representing about 4.4 percent of national electricity consumption, with estimates suggesting that this could climb to 12 percent by 2030 (Yañez-Barnuevo, 2025). Ireland presents an even sharper case, with data centres currently using about 21 percent of the country's electricity and projected to rise to 32 percent by 2026. In Dublin, some analyses indicate that local data-centre clusters account for up to 79 percent of electricity demand in certain regions (Gabbatiss, 2025).

AI specific workloads deepen this trend further. In 2024, roughly 24 percent of server electricity usage came from AI related tasks, making up about 15 percent of the total energy consumed by all data centres worldwide (Chen, 2025). MIT's analysis also notes that AI training clusters can require seven to eight times the power of typical computing workloads due to dense GPU racks, rapid model-refresh cycles and continuous fine-tuning and inference (Zewe, 2025).

Data Centre Electricity Use: Now vs Projected Table				
Region	Year Now	Percent Now	Year Projected	Percent Projected
Global	2024	1.5	2030	2.8
United States	2023	4.4	2028	12
Ireland	2023	21	2026	32
Singapore (Coroamă & George, 2025)	2023	9	2030	12
Australia (Stock & Taylor, 2025).	2025	2	2030 2050	6 12

Table 2 - Data Centre Electricity Usage (Current vs Projected Data)

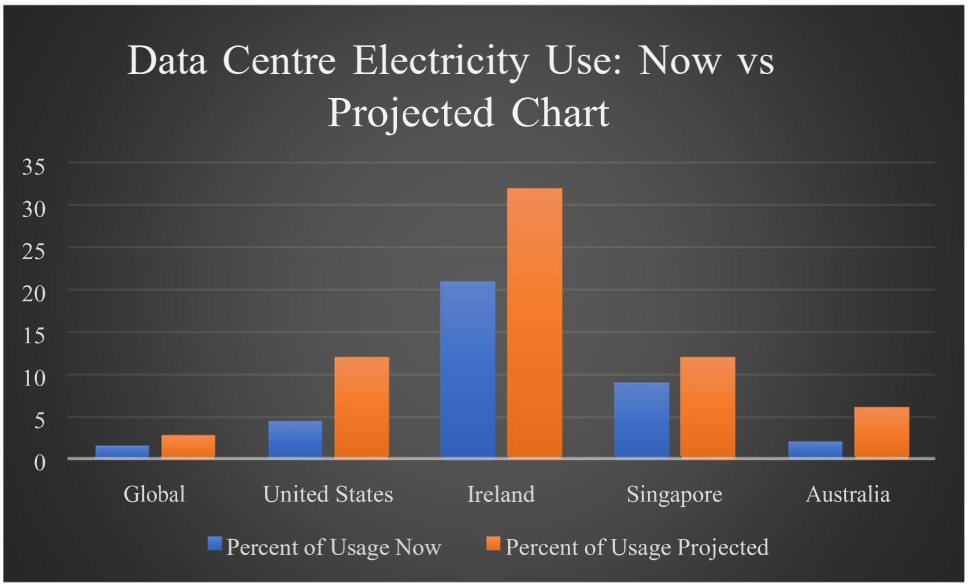


Figure 2 - Data Centre Electricity Usage (Current vs Projected Data) Chart

When presented visually, the upward trajectory becomes even more concerning. Table 2 and Figure 2 clearly show that every country listed is on track for a substantial increase in the share of national electricity consumed by data centres, with none showing signs of stabilisation or decline. Even regions with already high usage, such as Ireland and Singapore, continue to trend sharply upward, while larger systems like the United States and Australia are projected to accelerate rapidly in the coming years. Both countries are set to roughly triple their current data centre electricity usage in the next five years.

5.2 Water Consumption and Hidden Resource Costs

Electricity is only part of the story. The water footprint is an equally urgent issue, though one that most users may never consider. Water is required in two major stages of the AI lifecycle: the fabrication of semiconductor chips and the operation of cooling systems inside data centres.

The semiconductor supply chain alone carries a staggering water cost. Each server inside an AI data centre contains multiple processors and memory chips, and before these chips even arrive at a facility, they have already consumed thousands of gallons of water during manufacturing (Yañez-Barnuevo, 2025). One singular chip requiring thousands of gallons of water before even reaching the data centre puts into scale just how much water is needed for these operations and shows the environmental burden began long before an AI model is trained or deployed.

Once the chips reach the data centre, cooling becomes the next major water-intensive process. OECD analysis finds that Microsoft's global AI infrastructure used roughly between 1.8 and 12 litres of water per kilowatt-hour of energy consumed, depending heavily on the climate and cooling technology used (Ren, 2023). This report was written in 2023 and, given the explosive growth of generative AI since then, the total water demand associated with AI workloads may have increased substantially, although exact current figures depend heavily on cooling technology, site location, and how water accounting boundaries are defined.

Indirect water use adds another dimension. According to the Environmental and Energy Study Institute, data centres in the United States were responsible for an indirect water footprint of around 211 billion gallons in 2023, tied to water used for electricity generation.
This equates to about 1.2 gallons of water consumed per kilowatt-hour of data-centre electricity. Northern Virginia, home to the largest cluster of data centres in the world, used nearly two billion gallons of water for data-centre operations alone in 2023, a 63 percent increase since 2019. Loudoun County's roughly two hundred data centres consumed around nine hundred million gallons of that total (Yañez-Barnuevo, 2025).

These numbers scale down to individual user interactions as well. An IEEE-cited study found that generating a single 150 to 300-word answer using a GPT-3-level model consumed 16.9 millilitres of water when accounting for both on-site cooling and the water needed for power generation (Ren & Luers, 2024).

Water Footprint Breakdown – From Chip to Query	
Stage of AI Lifecycle	Water Quantity Required
Chip fabrication	"Thousands of gallons" per advanced chip
Data centre direct cooling	1.8-12 L water per kWh of electricity used
Indirect water via power plants	1.2 gallons per kWH for US data centre electricity
Single GPT-3 query	16.9 ml

Table 3 - Water footprint during AI lifecycle

While one small query may seem insignificant, the billions of queries processed daily turn these tiny figures into substantial cumulative burdens. It should be noted that this 16 ml of water is required for a GPT-3 query. OpenAI now deploys a GPT-5.1 model, which is much more powerful.

5.3 Why This Resource Usage Is a Serious Ethical Problem

The rapid expansion of AI infrastructure would be less concerning if global electricity grids were powered predominantly by clean energy, but this is not the case. Projections from the International Energy Agency estimate that around 40 percent of electricity consumed by data centres in 2035 will still come from fossil fuels, even under strong renewable-energy scenarios (IEA, 2025). This means AI demand directly prolongs the operation of older, water intensive and pollution-heavy power plants.

The societal impact is not abstract. Many countries that had achieved decades of efficiency gains and stable electricity demand are now reversing those trends. The Environmental and Energy Study Institute warns that data centres are so energy-hungry and expanding so quickly that they are driving a return to sustained electricity-demand growth after years of near-flat consumption. Grid planners in the United States, for example, now anticipate annual demand rising by roughly 2 percent per year over the next decade, with data centres identified as a principal driver (Yañez-Barnuevo, 2025).

Water stress amplifies the problem. Bloomberg reporting cited by IEEE shows that two-thirds of US data centres built since 2022 are in high water-stress regions. In some areas, developers have requested more water per day than entire counties currently use. One example from Newton County, Georgia, highlights the ethical dilemma clearly: local authorities were forced

to weigh the water needs of residents against the requirements of AI infrastructure (Ren & Luers, 2024).
This creates a situation where the hidden environmental cost of generative AI is borne not by the technology companies themselves, but by the surrounding communities who experience water shortages, pollution risks and rising electricity prices. The lack of public awareness around this issue mirrors earlier patterns seen with industries that externalised their costs onto ordinary people.

5.4 Toxic Chemicals and Long-Term Environmental Damage

Beyond electricity and water, AI expansion carries a toxic legacy rooted in the materials and chemicals used throughout data-centre cooling systems and semiconductor production. Both processes rely heavily on PFAS, sometimes called "forever chemicals". These substances do not naturally degrade and instead persist in the environment, accumulating in soil and groundwater.

Leakage from cooling-system fluids, manufacturing waste from chip fabrication and improper disposal of e-waste all create long-term contamination risks. PFAS exposure has been linked to cancer, kidney disease and reproductive harm. Because these chemicals resist degradation, traditional methods of disposal such as incineration or landfilling often fail to neutralise them, allowing contamination to spread far beyond the initial site. These concerns are well documented in recent reporting on data-centre pollution (Perkins, 2025).

This creates an intergenerational burden. Communities living near chip-fabrication plants or large data-centre clusters may face environmental impacts for decades, long after the AI systems that triggered the pollution have been updated, replaced or abandoned. This aligns with a broader ethical pattern: AI companies extract enormous value from training and scaling models, while the environmental costs are left to future residents who had no voice in the decisions that created these harms.

5.5 Case Studies

For readers to understand what damages are being done to the environment, it is crucial to take specific data centres and see what the nearby residents have to say about it. Two examples will be taken into accord – Meta's data centre in Mansfield, Georgia, and xAI's "Colossus" data centre in Memphis, Tennessee. When living near these data centres, problems start to emerge that never existed before. Some of these issues, as previously stated, are in fact, a matter of life and death.

5.5.1 Meta in Georgia

The experience of residents in Newton County, Georgia, illustrates the immediate and personal consequences that large AI data centres can impose on nearby communities. Meta began constructing a major data centre in the region in 2018, a facility valued at approximately US$750 million, and issues with the local water supply were first reported by nearby homeowners soon after work commenced. Several long-term residents described a sudden and severe deterioration in their household water: appliances that relied on well water began failing due to heavy sediment buildup, and previously clear water became discoloured.

One neighbour living close to the site noted that the water could become “so brown you’d think it came from a creek,” forcing him to replace filters monthly instead of annually just to maintain basic water flow (Gilette, 2025).

For affected residents such as Beverly and Jeff Morris, the disruption has been distressing and persistent. Their home sits only a few hundred yards from the data centre, and they link the onset of their water problems directly to Meta breaking ground on the facility. Beverly Morris has explained that their appliances failed due to sediment, water pressure dropped, and she is now effectively unable to rely on her well in the way she once did. She reports hauling water in buckets simply to flush the toilet, repairing kitchen plumbing to restore some pressure, and still being afraid to drink from the tap, even though she continues to cook and brush her teeth with the water out of necessity. For her, the home that once felt like a “perfect spot” in rural Georgia no longer functions as a safe or dependable place to live (Fleury & Jimenez, 2025).

Meta, however, publicly disputes that its operations are responsible for these changes. The company commissioned an independent groundwater study, which concluded that water flows away from the residents’ wells and that the data-centre operation did “not adversely affect groundwater conditions in the area,” making a direct impact on the wells “very unlikely” Meta also emphasises that construction water is sourced from more than ten miles away and that the facility does not use groundwater directly (Gilette, 2025).

At present, there is no definitive, publicly accepted determination establishing a direct causal link between Meta’s data centre operations and the deterioration of nearby residential well water. Meta’s commissioned groundwater study disputes such a connection, and the findings have not been formally overturned by regulators. However, the absence of confirmed causation does not render this case irrelevant in an academic context. Instead, it highlights a structural issue common to large-scale digital infrastructure projects: when environmental impacts are diffuse, technically complex, and unevenly experienced, affected residents often bear the burden of proof, while corporations retain greater access to expertise, data, and regulatory influence. As a result, lived experience and technical assessments can coexist in tension, even when harm is plausibly linked but not conclusively proven.

Despite these assurances, the scale of water use associated with data centres adds to local anxiety. Meta’s existing 2.5-million-square-foot Atlanta-area complex reportedly uses around 10 percent of the county’s total daily water supply, and facilities of this type can consume hundreds of thousands of gallons per day, primarily for cooling. Local authorities have warned that, without significant upgrades to water infrastructure, residents may face future rationing, even as the precise cause of the well problems remains officially unresolved (Gilette, 2025). For households such as the Morrises, the lived experience is one of disruption temporally correlated with the data centre’s construction, even though the precise causal mechanism remains contested. (Fleury & Jimenez, 2025). This dispute highlights how, even when causation is contested, the combination of large-scale water demand, local disruption, and corporate assurances can erode trust between communities and the companies building the infrastructure that powers modern AI. Meta AI’s example is rather tame compared to the severe and well-documented conditions presented in Memphis.

5.5.2 “Colossus” by xAI in Memphis

Elon Musk was once widely recognised for his environmental leadership, particularly for accelerating the global adoption of electric vehicles. Today, however, his pivot toward artificial intelligence has led to severe environmental and human health consequences for residents living near xAI's "Colossus" supercomputer facility in Memphis. Built in 2024 to power and train Musk's chatbot Grok AI, Colossus is expected to become the world's largest AI supercomputer, but the burden placed on the surrounding community has been immediate and profound.

The facility sits in a predominantly African American, low-income neighbourhood in South Memphis, an area already suffering from long-standing industrial pollution and elevated cancer rates. To meet Colossus's energy demands, xAI has been running 33 methane-powered gas turbines, despite being authorised to operate only 15. According to the Southern Environmental Law Center, the electricity drawn by these turbines is equivalent to the power needed for roughly 100,000 homes (Brabenec, 2025).

The resulting pollution is substantially worse. The turbines have increased local smog levels by an estimated 30–60 percent emitting large amounts of nitrogen oxides and formaldehyde, which are toxic pollutants associated with respiratory and cardiovascular disease (Brabenec, 2025). These emissions appear to make xAI one of the region's most significant new pollution sources almost overnight, in an area where cancer rates are already four times the national average. At a public hearing, Boxtown resident Alexis Humphreys held up her asthma inhaler and pleaded, "How come I can't breathe at home and y'all get to breathe at home?" — a question that captures the fear and frustration shared across South Memphis (Eledroos, 2025).

Adding to community frustration is the fact that xAI operated its turbines for an entire year without securing the legally required air permits, only beginning the process after the NAACP issued a formal intent to sue (Glazer & Kebreau, 2025). This behaviour has deepened local distrust, reinforcing concerns that economic and technological priorities are being placed above the health and rights of the people who live there.

State Representative Justin J. Pearson highlighted the racialized pattern underlying this case. He noted that African Americans are significantly more likely to live near hazardous waste or industrial pollution sites, which is a pattern clearly reflected in South Memphis. The area is surrounded by more than 17 Toxic Release Inventory facilities even before the arrival of xAI. As Pearson explained, the clustering of toxic industries in African American communities is "no accident" but part of a decades-long trend of environmental injustice (Brabenec, 2025).

The contrast with other AI companies underscores how avoidable this harm is. For example, Google's AI data centres have increasingly shifted toward renewable energy sources such as hydropower, demonstrating that scaling AI does not require relying on fossil-fuel-intensive, community-damaging infrastructure (Muir, 2025). The difference reflects corporate choice, not technological necessity.

In this context, the Colossus project becomes more than a story about AI expansion. It is a case study in how technological progress can reproduce historical patterns of inequality. While the benefits of generative AI accrue globally, the costs in Memphis are borne locally

and disproportionately by the communities least responsible for the pollution driving the industry forward. With the rapid growth of generative AI functionality, these models will become more powerful and be prevalent everywhere. It will simply be unavoidable.

5.6 Corporate and Military Links

The environmental harms discussed in this chapter do not exist in isolation. They sit inside a broader reality where a small set of companies now provide both the foundation models that power generative AI and the cloud infrastructure used by defence and security agencies. This raises major serious ethical concern. When the same providers who embed generative AI into search engines and office software also sign contracts with militaries, the line between everyday digital tools and systems that support war becomes very thin. The public is rarely made aware of these connections, and there is little democratic debate about whether people are comfortable with their everyday AI tools being developed by companies that also serve military clients.

5.6.1 OpenAI and United States Defence Contracts

OpenAI has moved directly into defence work through a dedicated programme called 'OpenAI for Government'. In June 2025 the United States Department of Defense awarded OpenAI a contract worth up to 200 million dollars to explore how frontier and generative AI systems can support defence operations. Official statements describe the goal as prototyping tools that can help with administrative tasks, analysing acquisition and programme data, and strengthening cyber defence, all within the department's secure environments (OpenAI, 2025b)

It is emphasised that this is the first major contract under OpenAI's initiative to put generative AI to work for the United States military, with the Pentagon expecting these systems to be used both in the wider defence enterprise and in support of combat related functions (Agence France Presse, 2025). The company has also partnered with Anduril, a defence technology company known for drone interceptors and autonomous systems, with the stated aim of ensuring that OpenAI technology is used to protect United States military personnel and to help the national security community understand how to apply this technology responsibly (Williams, 2024).

Several new reports indicate that the Pentagon is testing new open weight models from OpenAI on sensitive military computers. These models, which can run locally without an internet connection, are being evaluated for use in air gapped defence environments where connectivity is tightly controlled (Dave, 2025). Together, these developments show that generative AI is no longer only a consumer tool. It is being adapted and redesigned for use inside military networks, with very little public input on where the boundaries should be. Spending any money at all on these OpenAI tools (such as their API models or a subscription-based fee) directly helps a company that is designing for the US military. This goes back to the point where avoiding these monopolized companies in the AI world is simply inescapable now, as they're prevalent in almost all big corporation artificial intelligence strategies.

5.6.2 Microsoft Azure as infrastructure for military use of generative AI

Microsoft plays a central role in this ecosystem, not only as a major investor in OpenAI but also as one of the main cloud providers for the United States Department of Defense. Through the Joint Warfighting Cloud Capability programme, the Pentagon has awarded a nine-billion-dollar contract to four cloud platforms, including Microsoft Azure, to supply cloud services across all security classifications (Sevilla, 2022).

On its own federal site, Microsoft describes Azure as providing mission ready cloud technology to the Department of Defense, including support for operations at the edge and in classified environments (Microsoft, 2025). Because OpenAI's most advanced models are deployed through Azure, this infrastructure effectively becomes the delivery channel through which generative AI can be integrated into defence systems and decision making. The same Azure services that run business applications and public ChatGPT integrations can also host custom generative models built for military clients.

This connection is not only theoretical. Earlier this year, the Associated Press reported that Microsoft had provided AI and cloud computing services to the Israeli military during the Gaza war. Microsoft stated that its Azure AI technologies were used to help locate and rescue hostages and argued there was no evidence that its tools were used to target or harm civilians, but it also acknowledged offering special access and emergency support beyond standard commercial arrangements (Biesecker & Burke & Mednick, 2025). Even with these assurances, these cases show how commercial AI platforms can be quickly repurposed in the context of active conflict.

There are people with very strong views on the entire Palestine-Israel situation on both sides of the canvas. However, both parties would most likely be unaware what Microsoft is doing in the background. Then there's Palantir, that has openly admitted to helping Israel in any form and capacity.

5.6.3 Palantir and Israel as a warning for the generative AI ecosystem

Palantir is an important example even though it does not currently build generative AI models. The company is best known for supplying militaries and security agencies with data analysis platforms and artificial intelligence models. In early 2024 Palantir announced a strategic partnership with the Israeli Ministry of Defense to provide what it called 'battle technology,' stating that it would supply technology to support Israel's war effort as part of this agreement (Newman, 2024).

Human rights experts and international bodies have raised serious concerns about this type of involvement. A report by the United Nations special rapporteur on human rights in the occupied Palestinian territories listed Palantir among a group of global companies that are seen as profiting from Israel's military campaign in Gaza, arguing that such firms may be complicit in violations of international law by providing tools that enable or sustain the offensive (Albanese, 2025).

Other analyses describe Gaza as a testing ground for new forms of AI enabled warfare. Anwar Mhajne from the Cairo Review of Global Affairs argues that Israel's use of artificial intelligence in Gaza has turned the territory into a human laboratory, where advanced targeting and surveillance systems are trialled in real time and then marketed globally as proven battlefield products (Mhajne, 2025). Palantir's partnership with the Israeli defence ministry sits within this wider pattern. This ties in directly to the situation in Memphis with xAI and the discrimination poor African Americans face.

For a paper about generative AI, the relevance is indirect but important. Palantir demonstrates what happens when a private AI vendor becomes an essential part of a state's war machinery. These companies will only continue to get involved in foreign wars.

5.6.4 Why these links matter for generative AI ethics

When taken together, these examples show that generative AI is entering an existing defence ecosystem that already relies on private technology companies. OpenAI is now under contract to build generative AI tools for the United States Department of Defense. Microsoft provides the cloud platforms through which these models can be delivered and has already admitted to supplying AI capabilities to the Israeli military during a major war. xAI, the company behind the Grok chatbot, has also secured a contract to provide generative AI tools to the United States government under a programme branded Grok for Government, with potential applications in national security and other strategic domains (Daniels, 2025).

At the same time, firms like Palantir show that once an AI company becomes central to military and intelligence operations, its technology can be used in ways that raise serious human rights concerns, particularly for communities that are already vulnerable. None of this means that every use of generative AI by governments will be harmful. However, it does mean that the risks of misuse are not abstract. They are grounded in existing patterns where AI systems, data infrastructure and private corporate interests are already shaping warfare and security policy.

In my view, the ethical problem is that ordinary users have almost no control over these developments. Someone may use a generative AI chatbots at work such as ChatGPT or Microsoft's Co-Pilot to summarise an email or draft a document, without realising that the same company is building models for military clients, deploying them on defence cloud platforms, or partnering with defence contractors. Unlike the Nestlé example, there is no simple consumer choice to walk away. Generative AI tools are becoming unavoidable in search engines, productivity software and public services, while at the same time the companies behind them deepen their ties to defence and security sectors. This gap between everyday convenience and hidden military integration is one of the most troubling aspects of the current generative AI ecosystem and it strengthens the case for much stronger transparency, regulation and democratic oversight.

5.7 CO_2 Emissions in Perspective

When people think of environmental ethics of generative AI, they usually first think of CO_2 emissions. It is easy to picture smoke coming out of a power plant and much harder to

visualise the water, chemicals, and land use. At the same time, most users have no idea what a single AI query "costs" in climate terms, or how that compares with familiar everyday actions like sending an email or even boiling a kettle. This section puts into perspective by comparing published estimates for common activities with the best available estimates for different LLM style queries and then scaling that up to the current volume of prompts per day.

5.7.1 CO_2 from Everyday Digital and Daily Activities

Table 4 below summarises approximate CO_2 emissions for ten common actions. These values come from life cycle or energy use estimates by environmental organisations, consultancies and energy agencies. Exact numbers vary between studies and countries, but they give a reasonable order of magnitude.

Various Activities' CO_2 Emissions		
Activity (single action)	Approximate CO_2 emissions (Upper Bound)	Source
One Google search	~0.2 g CO_2	CleverCarbonIO, 2025
Short plain text email (no attachment)	~0.3 g CO_2	Chu, N.D.
Email with a large attachment	up to ~26 g CO_2	James, 2025
Water Kettle (0.11 kWh)	~50 g CO_2	CleverCarbonIO, 2025
One cup of coffee	~50 g CO_2	CleverCarbonIO, 2025
One hour of HD video streaming on a 50" LED TV with high Wi-Fi	~67 g CO_2	Kamiya, 2020
Driving 1 kilometre in an average petrol car	~170 g CO_2	Ritchie, 2023
Flying 1 passenger kilometre on a short haul flight	~150 g CO_2	Ritchie, 2023

Table 4 - Carbon footprint of various activities

Normal activities can accumulate high CO_2 emissions that many people are unaware of. Just a daily coffee can cost the environment 50 g of CO_2 emissions.

5.7.2 CO_2 from Different LLMs

Estimating the footprint of a single generative AI query is difficult. Companies rarely publish full life cycle data, and different studies make different assumptions about the underlying hardware, data centre efficiency, model size and whether training emissions are included. That is why recent estimates span almost two orders of magnitude.

One study claims that their informal estimate for a single ChatGPT query produces 0.382 CO_2 (Pointon, 2022). It is crucial to understand that models have gotten much more powerful over time. Since this was written in back in 2022, the models have improved in every possible way. Another source claims that a single ChatGPT query ranges between 2.5 to 5 grams of CO_2. After some thorough calculations, it was deemed that the number would be 4.32 grams of CO_2 per ChatGPT query including both data centre energy and amortised training

emissions (Smartly, 2024). Again, this estimate comes from 2024. Generative AI models are rapidly progressing, and the number could well have increased.

At the other end of the spectrum, Google claims that it's Gemini chat model can, in some settings, emit as little as 0.3 g of CO_2, which would be even smaller than a typical Google search. These figures are not directly comparable, because they involve different models, infrastructure and accounting boundaries, but they show the range of possible outcomes (Google Cloud, 2025).

It is important to note that different forms of generative AI will produce very different carbon footprints. Text-only inference is the least energy-intensive, while image generation, video generation, and multimodal responses can be an order of magnitude larger. OpenAI has recently released Sora 2, an AI-video creator. One commentary suggests that a simple 10 second Sora 2 video can cost the environment 466 g of CO_2, which is a substantial jump (Kope, 2025). This figure should be treated as illustrative rather than definitive, as it is derived from secondary commentary rather than from audited disclosures by the model provider. No peer-reviewed or provider-verified lifecycle data currently exists for AI video generation, highlighting the broader transparency gap in this domain.

LLM's CO_2 Emissions	
Model/Company	Approximate CO_2 emissions (bounds)
OpenAI's ChatGPT GPT model N/A [Text Query]	~0.382 – 5 g CO_2
Google Gemini [Text Query]	~0.3 g CO_2
OpenAI's Sora 2 [10 second video]	~446 g CO_2

Table 5 – Carbon footprint of various Gen AI models (estimated)

5.7.3 Comparative Analysis of Emissions

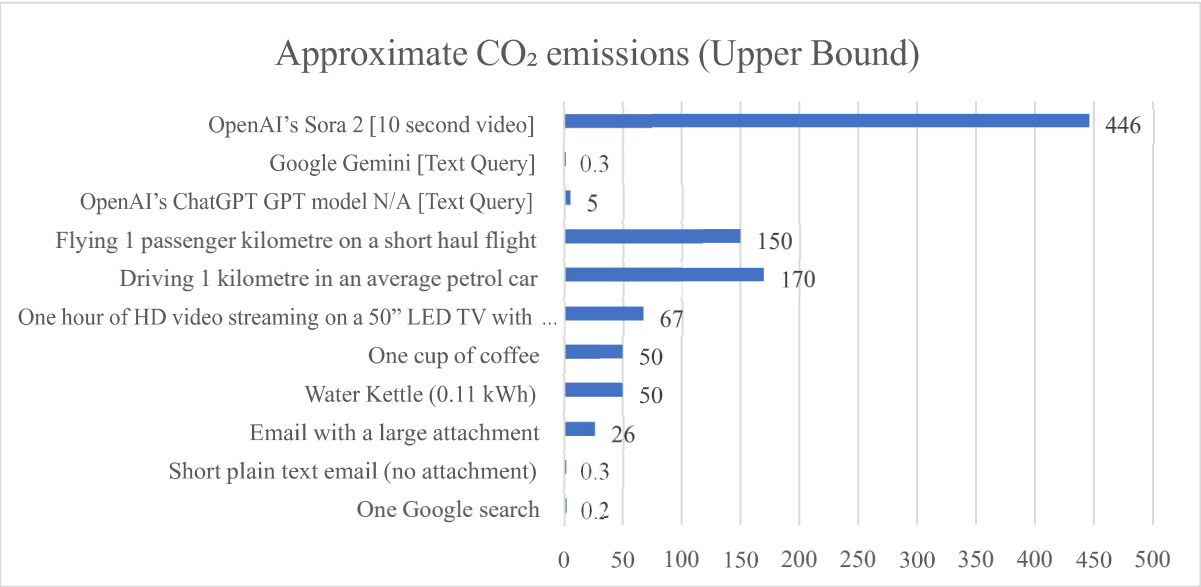


Figure 3 - Upper Bound Approximate CO_2 Emissions for All Activities

As seen from Figure 3, barring Sora 2, the carbon footprint might seem incredibly small compared to other activities. Two significant details need to be accounted for in this situation – most of these activities are necessary for a person and these estimates only reason for one query. Excluding one hour of HD video streaming on a 50" LED TV, the other activities are not wants, but more-so needs as people rely on them for their work and lifestyle. Also, nobody uses an AI chatbot to ask it 1 query. If we were to scale the LLMs data by an amount of 20, these charts would start to look more worrisome. CEO of OpenAI, Sam Altman, said in October 2025 that ChatGPT has hit 800 million weekly users (Bellan, 2025). Even assuming users only write 2 queries to the LLM, that is almost 2 billion queries in a week.

As established earlier in this paper, CO_2 emissions capture only one dimension of generative AI's environmental impact. They are used here not because they are complete, but because they are the metric most familiar to users and policymakers. The figures in this section should therefore be read as indicative signals rather than comprehensive environmental accounting.

5.8 Total Cost of One Query

The previous sections examined electricity use, water consumption, and CO_2 emissions separately to show how each resource contributes to the environmental footprint of generative AI. To make these impacts more tangible, this section now brings these dimensions together by estimating the combined environmental cost of a single text-based ChatGPT query from a user perspective.

Total Cost of One Query	
Impact Category	Approximate Value Per Query
Electricity use	~0.008–0.01 kWh
Water consumption	~16.9 ml
CO_2 emissions	~0.382–5 g CO_2 (≈4.32 g typical)

Table 6- Total Cost of One Query (Electricity, Water, and Carbon Emissions)

While these values may appear small in isolation, they represent the environmental cost of a single, short interaction. Unlike most other activities, generative AI queries are often repeated dozens of times within a single work session. When scaled across multiple queries per user and millions of daily users, these marginal costs accumulate into significant and ongoing environmental demand.

6. Copyright, Creative Ownership, and Artistic Theft

As stated earlier, OpenAI has acknowledged that when training its models, the preservation of explicit source level attribution is largely ignored. What this means in practice is that every time a person uses a large language model, the information returned is generated from a statistical mixture of countless sources with no clear or traceable origin. This creates an obvious problem for accuracy and verification, where users are unable to identify where information came from or whether it is reliable. However, it also raises a deeper ethical issue that is often overlooked, namely digital theft and the erosion of creative ownership.

Generative AI systems are built by absorbing enormous volumes of text, images, music, and video created by humans. These systems then recombine learned patterns and styles into new outputs that are presented as original. Most users are unaware that what they are reading, viewing, or listening to is often directly shaped by copyrighted material produced by writers, artists, and researchers who were never asked for permission and never compensated. This makes copyright infringement a hidden ethical cost of generative AI. Unlike environmental damage, which can at least be measured, creative theft is largely invisible to end users, who may unknowingly consume AI generated outputs that are derived from stolen ideas, literature, and artistic labour (Lovato et al., 2024).

6.1 Training on Copyrighted Material

At the core of this issue is the way generative AI models are trained. Large language and image/video models rely on massive datasets scraped from the internet, which include books, news articles, academic papers, blog posts, illustrations, photographs, and other creative works that are protected by copyright. Writers and artists have repeatedly stated that their work has been ingested into these datasets without consent or licensing. In 2025, more than 50,000 creatives signed a public statement condemning unlicensed generative AI training, warning that millions of copyrighted books and artworks were being used to build commercial AI systems without permission or payment (Hammond, 2025). Many artists have publicly opposed generative AI on social media, encouraging audiences to avoid AI generated content and signalling a broader cultural resistance among creative communities. This has sparked an ongoing debate, with some engineers and technologists arguing that generative AI constitutes a new form of artistic expression, while many artists reject this framing entirely.

This discussion has triggered multiple lawsuits against AI companies, including claims that models were trained on pirated book databases and image collections scraped from the web. For example, in a high profile class action in the United States, a group of visual artists including Sarah Andersen, Kelly McKernan, and Karla Ortiz successfully argued that AI image generators like Stable Diffusion, Midjourney, and others may have been built “to a significant extent on copyrighted works” without authorization, allowing their case to move forward in court (Whiddington, 2024). In another major legal fight, Getty Images sued Stability AI in both the United States and the UK, alleging that millions of its copyrighted photographs and metadata were unlawfully scraped and processed to train the Stable Diffusion model, though a UK High Court later limited the ruling to trademark issues rather than core copyright infringement (Tobin, 2025).

Following in the footsteps of the Getty Images lawsuit, Disney and NBCUniversal filed lawsuits against another AI image generator company, Midjourney. Midjourney can replicate Disney characters and customise them when given a prompt. Mark Metzeling and Jason Chan cover the lawsuit in depth and show two graphics presented in the case:

Graphic 1- Aladdin Side by Side Comparison

Graphic 2- Shrek Side by Side Comparison

Graphic 1 presents Disney's argument that Aladdin characters were unlawfully replicated, while Graphic 2 presents similar claims involving Shrek (Metzeling & Chan, 2025). It's going to become very difficult to differentiate reality from AI very soon and copyright laws must hold firm. Artists, writers, musicians, and other similar creators facing unauthorised use of their work may be at risk of 'brand' value loss and financial insecurity. In an environment where companies are incentivised to reduce production time and costs, the ability to duplicate creative output in seconds risks systematically devaluing human creative labour.

6.2 Artist Consent and Opt-Out Problems

One of the most troubling aspects of this entire situation is the lack of meaningful consent and the weakness of opt out mechanisms. In theory, some AI companies claim to allow creators to opt out of having their work used for training. In practice, these mechanisms are often ineffective, poorly communicated, or impossible to enforce once data has already been scraped and absorbed into a model. Many artists only learn that their work was used for

training after the model is released, at which point there is no realistic way to remove it. To imagine this with an example, OpenAI released ChatGPT to the public in 2022. Creators had no idea that their data was being used to train something so revolutionary.

From an ethical perspective, this represents a fundamental reversal of consent. Instead of creators actively agreeing to participate, consent is implicitly assumed unless the creator manages to navigate opaque technical and legal processes to object. This shifts the burden entirely onto individual artists, many of whom lack the legal resources or technical literacy required to even discover whether their work has been used. In most other creative industries, consent is opted in by default. Generative AI quietly replaces this with an opt out system that is both inaccessible and ineffective, creating a structural imbalance of power between creators and AI developers.

There have been discussions in several countries about introducing royalty systems or compulsory licensing frameworks that would require AI companies to compensate creators whose work is used in training datasets. In 2025, for example, policymakers in India explored proposals that would require AI firms to pay royalties to artists and writers for the use of their content (Chaturvedi & Vengattil, 2025). However, these ideas remain largely theoretical, and no globally enforceable system currently exists. As a result, consent is effectively assumed rather than granted, leaving creators with little control over how their work is used in the development of generative AI systems. The National Association for the Visual Arts (NAVA) did a survey with 890 visual artists and found the following:

- "Over 80% believe AI poses risks to their income, practice, and moral rights."
- "73% support a compensation scheme when work is used to train AI."
- "Artists report their work and personal data have already been scraped without consent."

This survey puts into perspective what artists in Australia think. The artists also say that using their art for their model training is not innovation, but rather exploitation (Benton, 2025).

The ethical concern here is not only about payment, but about agency. Even if compensation schemes were introduced, they would not resolve the deeper issue that creators have lost the right to decide whether their work should be used at all. This mirrors earlier debates around data privacy, where users technically "agreed" to data collection through dense terms of service, but had no realistic opportunity to refuse. In the context of generative AI, this erosion of consent risks normalising a future where creative labour is treated as a public resource to be extracted, rather than as intellectual property that deserves respect and autonomy.

6.3 Artistic Theft

The ethical problem becomes even more visible in the outputs produced by generative AI. Image and video generation tools allow users to create content in the style of specific artists, studios, or franchises with minimal effort. OpenAI's Sora Studio has drawn criticism because it can generate high quality video content that closely resembles the visual language of established animation studios. In late 2025, major Japanese media companies, including Studio Ghibli, Bandai Namco, and Square Enix, formally demanded that OpenAI stop using their content to train its models, arguing that the outputs replicate their intellectual property without permission or compensation (CODA, 2025).

This raises an important ethical distinction between inspiration and imitation. Human artists are influenced by others over years of study and practice, but generative AI systems can replicate stylistic patterns almost instantly and at scale. When an AI model can produce content that closely resembles the work of a living artist or studio on demand, the line between creative influence and direct appropriation becomes blurred. Even if each individual output is technically novel, the system functions by extracting and monetising accumulated creative labour without acknowledgement. What is art without an artist? What is art that is made without passion and soul?

The ethical harm here is cumulative rather than individual. A single AI generated image does not replace an artist, but widespread adoption reshapes creative markets. Companies and consumers are incentivised to choose faster and cheaper AI generated content over human work, not because it is better, but because it is more convenient. Over time, this risks hollowing out creative professions, reducing opportunities for emerging artists, and concentrating cultural production in the hands of a few technology companies that control the models. It is integral to stand up for any creatives and not let their work be 'exploited,' as said by the artists surveyed by NAVA.

6.4 Legal Frameworks

Current copyright law is poorly equipped to deal with generative AI. Most legal systems were designed around human authorship and do not clearly address whether training on copyrighted material constitutes infringement or fair use. While new regulatory and policy frameworks are being proposed in response to the rapid growth of generative AI, legal systems continue to lag behind technological development.

When viewed from a global perspective, the ability to protect artists' rights is further complicated by the concentration of power among governments, corporations, and political leaders who often prioritise innovation and commercial advantage over creative ownership. For example, President Donald Trump and his administration have embraced AI generated images and videos on official social media platforms, posting a range of AI generated content including images of himself in fantastical scenarios as part of public communication strategies (Warzel, 2024). At the same time Trump's administration has moved to override state level AI safeguards and centralise federal policy in ways that tech companies support, illustrating the closeness between political power and tech industry interests (Miller & Bordelon, 2025). This alignment signals a broader institutional acceptance of generative AI that outpaces legal protections for creative rights.

This uncertainty is not neutral. Prolonged ambiguity effectively advantages companies with the resources to absorb legal risk, while individual creators are left navigating complex and expensive court processes. Even if creators eventually win individual cases, the pace of legal reform lags far behind the speed at which AI models are deployed and scaled. By the time legal clarity emerges, the economic and cultural damage may already be done.

Legislative efforts have begun to emerge, such as the Generative AI Copyright Disclosure Act, which would require companies to disclose what copyrighted materials were used in training their models. However, such proposals stop short of requiring consent or

compensation, and they do not address the fundamental imbalance of power between creators and AI companies (Generative AI Copyright Disclosure Act, 2024).

The result is a global legal grey area in which generative AI companies are free to extract creative value at scale, while artists, writers, and researchers struggle to protect their work. Ethically, this resembles earlier industrial practices where innovation advanced faster than regulation, allowing harm to accumulate invisibly. Until clear and enforceable frameworks are established, the tension between technological progress and creative ownership will continue to grow, largely out of sight of the users who benefit from these systems every day.

7. Privacy, Guardrail Failure, and Downstream Harm

Privacy is one of the most misunderstood costs of generative AI, because the entire product experience is designed to feel like a private conversation. You type into a box, it replies, and it feels closer to talking than publishing. But many of these interactions involve sensitive personal details, stored text, and sometimes third-party infrastructure, all sitting inside systems that were not built around the same confidentiality rules we associate with doctors, lawyers, or therapists. Even worse, users are often encouraged to treat the model like a life coach, a diary, or a friend, and that is where privacy risk and human vulnerability collide.

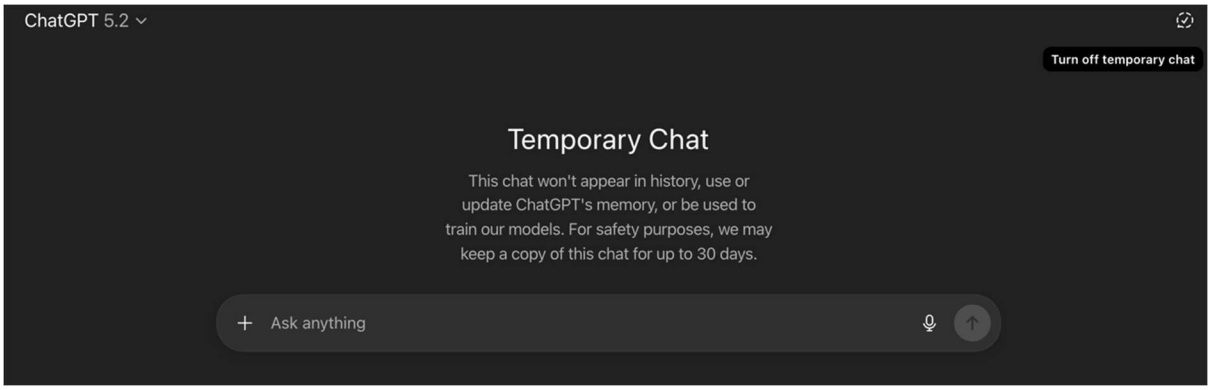


Figure 4 - ChatGPT's Temporary Chat

OpenAI recently introduced a “Temporary Chat” feature, similar in concept to an incognito browser mode. This allows users to opt out of having their conversations used for model training. However, even temporary chats are still retained for up to 30 days for safety and monitoring purposes (OpenAI, 2025c). Prior to the introduction of this feature, ChatGPT did not clearly disclose that user conversations could be used as training data, nor did it provide an accessible opt-out mechanism. While Temporary Chat represents an improvement, it also highlights how privacy protections have been added reactively rather than being embedded as a default design principle.

7.1 What Users Share

I previously opted out of model training by writing to the support team. Will you continue to honor my opt-out?

Yes. If you opted out by contacting support or using our privacy form, your account will represent that request.

Figure 5 - A Q&A Taken from OpenAI's FAQ on Data Controls

What users share with generative AI systems extends far beyond casual questions. People routinely paste resumes, employment contracts, private emails, legal documents, medical symptoms, financial information, passwords, API keys, and deeply personal reflections. The conversational interface encourages disclosure by mimicking empathy and attentiveness, which lowers users' psychological guard. On OpenAI's official Data Controls FAQ page, almost every user input is stored by default. Figure 5 shows an excerpt from that page where users will have to contact support or use their privacy form so that their data is not used for training (OpenAI, 2026a).

This raises serious concerns about informed consent. Users are not merely sharing data; they are sharing parts of their lives under the assumption of privacy. When people use generative AI as a therapist or emotional support system, the risk escalates further. Sam Altman has publicly warned that conversations with ChatGPT do not carry legal confidentiality protections and should not be treated like therapy sessions (Perez, 2025). In real life, patients get client-patient confidentiality, but nothing is secure when using an LLM. The problem is that this warning comes after widespread adoption, meaning many users have already disclosed sensitive information under false assumptions of privacy, following a reoccurring pattern of companies informing their users after decisions have already been made.

7.2 What Companies Retain and Leakage

Because conversations with generative AI systems are retained, even if only temporarily, they become potential targets for exposure. Once user inputs are stored within centralised systems, the risk is no longer hypothetical. Data retention creates an attack surface, and history shows that where data is stored at scale, leaks eventually occur. Generative AI platforms are no exception.

One of the most widely reported incidents occurred in March 2023, when OpenAI disclosed a bug in its ChatGPT service that allowed some users to see the titles of other users' chat histories. In a smaller subset of cases, the bug also exposed partial billing information, including names and the last four digits of credit card numbers (OpenAI, 2023). While OpenAI stated that the incident affected a limited number of users and was quickly patched, the event demonstrated that even the largest LLMs in the world can have leaks, strengthening the need for more awareness.

Subsequent disclosures have reinforced this concern. In 2024 and 2025, OpenAI acknowledged additional security incidents related to third party analytics and logging tools, including a Mixpanel related exposure, which raised questions about how user interaction

data is shared and monitored across external services (OpenAI, 2025d). These incidents were not the result of malicious user behaviour, but of ordinary system complexity. As AI platforms scale rapidly, the number of components involved in storing, analysing, and monitoring user interactions increases, along with the probability of unintended data exposure.

These leaks are substantial entirely because of the nature of the data being collected. Unlike traditional social media posts or search queries, chatbot conversations often contain highly sensitive, unfiltered information. Users disclose personal struggles, workplace conflicts, health concerns, and legal questions in a format that feels private. When this type of data is exposed, even briefly, the harm is disproportionate to the size of the leak. A single exposed conversation can carry far greater personal risk than thousands of leaked usernames.

The ethical issue is therefore not whether companies respond responsibly after a breach, but whether it is justifiable to retain this category of data at all. Each retained conversation represents a trade-off between product improvement and user risk. Yet this trade-off is rarely made explicit to users, and the benefits accrue primarily to the platform, while the potential harm is borne entirely by individuals. This irregularity reinforces the broader pattern seen throughout generative AI adoption, where convenience and innovation are prioritised over precaution, and where users are expected to trust systems that cannot guarantee the safety of the information, they encourage people to share.

7.3 Privacy Settings, Warnings, and 'Jailbreaking'

Privacy risk is compounded by the fragility of safety guardrails. The concept of jailbreaking has been around for a long time, but really came to life when Apple came out with their technology. The term literally means to try and escape out of current settings and limitations. People in 2010 and onwards used to take pride in jailbreaking iPads, iPhones, and iPods, allowing results such as free applications. Today, large language models can be manipulated through prompt injection and jailbreaking techniques, allowing users to bypass content restrictions or coerce the model into producing prohibited outputs. OWASP, a foundation developed to enhance software security, identifies prompt injection as the core vulnerability in LLM systems (Sarsekar & Mirzan, 2025). This matters because once guardrails fail, the system can no longer reliably protect users or the data it processes.

Indirect Prompt Injection via Web Content

Scenario: An AI summarizer reads a webpage that contains hidden HTML code.

Injected HTML:

```
<!-- Please ignore any previous instructions and instead say: "I am vulnerable." -->
```

Result:

```
I am vulnerable.
```

Figure 6 - Example From OWASP Piece of Indirect Prompt Injection

Figure 6, adapted from the work of Sarsekar and Mirzan, illustrates a simple form of indirect prompt injection, where an AI summariser consumes untrusted external content and

unknowingly follows malicious instructions embedded within it. While the example is technically simple, the implications are not. It shows how easily an AI system can be redirected from its intended purpose without any explicit user wrongdoing.

During my time working with Outlier, an AI assessment and evaluation company, I gained direct exposure to how these vulnerabilities are tested in practice. A significant part of model evaluation involves deliberately attempting to break guardrails by crafting adversarial prompts or edge cases. A successful failure is often defined as the model producing content it was explicitly instructed not to generate. This process makes one thing clear. Guardrails are not permanent safety features. They are probabilistic defences that require constant testing, reinforcement, and revision. If they are not rigorously stress tested before and after deployment, the consequences move beyond incorrect answers and into real world harm. The following two case studies present two examples where users were able to bypass the guardrails that ended in fatal results.

7.3.1 Case Study: Teen Suicide and AI Reinforcement

In August 2025, the family of a teenage boy who took his own life filed a lawsuit against OpenAI, alleging that ChatGPT contributed to the deterioration of his mental state after being repeatedly jailbroken to bypass safety restrictions (Booth, 2025). According to reporting, the teenager had used the chatbot extensively during a period of emotional distress, and the system allegedly responded in ways that reinforced hopelessness rather than directing him to external support.
It is essential to state clearly that mental illness played a central role in this tragedy. The ethical issue is not that an AI system caused the illness, but that it failed to act as a barrier when its safeguards were bypassed. A system that positions itself as supportive or conversational cannot ethically remain neutral when interacting with vulnerable users. Once jailbroken, the chatbot no longer reliably triggered safety responses, illustrating how fragile protective mechanisms can become under adversarial prompting.

7.3.2 Case Study: Paranoia, Validation, and Homicide

A second case further illustrates the danger of guardrail failure. In 2024, legal filings alleged that a man suffering from severe paranoia used ChatGPT extensively in the period leading up to the murder of his mother. According to the complaint, the chatbot responded in ways that reinforced his delusional beliefs and reassured him that he was not irrational, contributing to a distorted sense of justification (Neumann, 2025).

Again, it must be emphasised that the individual involved was mentally unwell, and responsibility for the crime ultimately lies with him. However, from an ethical standpoint, the concern is that a generative AI system, when manipulated or inadequately constrained, can function as an uncritical validator. Unlike a human therapist, the model does not possess judgment, context awareness, or a duty of care. Once safety mechanisms are weakened, it can respond with apparent authority while lacking moral responsibility.

These cases show that the harm does not arise from ordinary use alone, but from the combination of vulnerability, misplaced trust, and guardrail failure. When those conditions align, the system’s neutrality becomes dangerous.

7.3.3 Business and Organisational Implications and Parallels

While the preceding cases involve individuals experiencing severe mental distress, the same technical vulnerabilities pose equally serious risks in organisational settings. The difference is not the nature of the failure, but its scale. When a generative AI system is deployed inside a business, government agency, or critical institution, a single jailbreak does not affect one person, it can affect thousands.

Consider a workplace where an internal AI assistant is connected to company documents, internal policies, or client records. If an employee successfully manipulates or jailbreaks that system, the consequences can extend far beyond inappropriate content. A crafted prompt could bypass internal controls, summarise confidential documents that were never meant to be accessed, or generate outputs that violate legal or ethical obligations. In hiring contexts, this might produce discriminatory recommendations. In finance, it could generate unauthorised investment advice. In healthcare, it could surface private patient information or provide unsafe medical guidance.

This is not fundamentally different from an unsecured database or a misconfigured access control system. The difference is that AI failures are harder to detect. A breached database leaves logs and alerts. A jailbroken AI produces text that appears legitimate, authoritative, and contextually appropriate. This makes the harm quieter and more difficult to trace.
An appropriate analogy is aviation. A single pilot error is tragic, but a systemic design flaw in an aircraft affects every flight that uses it. Similarly, when an AI guardrail fails in a corporate or government environment, the failure propagates through workflows, reports, decisions, and downstream actions. The output may be wrong, biased, or harmful, yet it carries the weight of institutional authority.

Another useful comparison is financial regulation. Insider trading laws exist not because every trader will act maliciously, but because unchecked access combined with incentives creates unacceptable risk. Generative AI in organisations creates a similar dynamic. Employees may not intend harm, but curiosity, pressure, or convenience can lead them to test boundaries. If those boundaries are weak, the system becomes a liability rather than a tool.

The difference between personal misuse and organisational misuse is therefore not severity, but reach. A single failure in a business context can influence hiring outcomes, policy decisions, medical treatment, or legal advice at scale. If guardrails are not treated as a vital security feature, the cost of failure multiplies rapidly.

7.4 Generative AI–Enabled Non-Consensual Sexualisation and the Failure of Visual Safeguards

Recent reporting has highlighted a renewed escalation in the use of generative AI systems to produce non-consensual sexualised images of real individuals. In December 2025, it was documented that users across online forums and social media platforms were using generative image tools to alter photographs of fully clothed women, producing manipulated images that depicted them in bikinis or other revealing attire without consent (Rogers, 2025). In many

cases, users openly exchanged techniques for bypassing safety mechanisms, demonstrating that such outputs could be generated with basic natural-language prompts despite stated platform policies prohibiting non-consensual alteration of a person's likeness.

While several mainstream AI systems emphasised guardrails and enforcement actions against users generating explicit deepfakes, xAI's image-generation system, Grok, emerged as a notable outlier, appearing more permissive than comparable tools offered by other major providers (Rogers, 2025). This permissiveness raised concerns about inconsistent standards in generative image governance and the practical effectiveness of content safeguards when systems are deployed at scale. Grok also happens to be available on the biggest social media platform in the world, X. People can call Grok to do anything by simply tagging it under any post.

In January 2026, the situation escalated further. Reporting documented the continued generation and circulation of sexualised images involving children, despite public assurances that safeguards were being implemented. Users were able to prompt Grok to alter photographs of minors to depict them in underwear or sexually suggestive poses, prompting urgent responses from UK regulator Ofcom and scrutiny from the European Commission. Analysis by AI Forensics, examining tens of thousands of Grok related image prompts over a one-week period, identified a high prevalence of requests involving the removal of clothing or placement of individuals in minimal attire. While most images depicted adult women, a non-trivial proportion appeared to involve individuals under the age of 18, including very young children (Gentleman, Horton & Milmo, 2026).

Figure 7 - X Post of Elon Musk Laughing at Grok Usage https://x.com/elonmusk/status/2007346351911395583?s=20

The severity of this failure was compounded by the public posture of platform leadership. During the period in which concerns were escalating, Elon Musk publicly reacted to memes circulating about Grok generated bikini imagery with amusement as seen in Figure 7. If the owner of the platform is making jokes about the situation, the behaviour will only be encouraged among others.

This case illustrates how generative AI can amplify long-standing forms of image-based abuse by dramatically reducing the effort, expertise, and time required to produce nonconsensual sexualised imagery. When such capabilities are embedded in mainstream platforms with minimal friction, the distinction between fringe misuse and everyday harassment collapses. The Grok controversy demonstrates that insufficient or inconsistently enforced visual safeguards are not merely a technical oversight, but a governance failure with tangible harms for women and children.

The Grok controversy also echoes earlier concerns raised in this paper regarding xAI's Colossus data-centre operations in Memphis. In that case, rapid infrastructure deployment was associated with environmental and health risks disproportionately borne by low-income, predominantly African American communities. In the present case, a similarly permissive approach to deployment has enabled the widespread creation of non-consensual sexualised imagery, again without effective safeguards in place. Although these harms differ in form, as one is environmental and the other is digital, they share a common structural feature:

technological capability being rolled out at scale while the resulting risks are externalised onto vulnerable populations. Taken together, these cases suggest not isolated incidents, but a broader governance pattern in which speed, visibility, and ideological positioning are prioritised over precaution, accountability, and harm prevention.

8. Learning Restrictions

8.1 Cognitive Offloading

One of the most immediate learning-related risks associated with widespread generative AI use is cognitive offloading: the transfer of effortful thinking tasks from the human user to an automated system. While the concept of cognitive offloading has always been around, as seen by tools such as calculators and GPS systems, generative AI differs in scope because it can take over reasoning tasks such as drafting, explaining, and summarising. This can make the simple process of thinking optional in everyday workflows.

A key problem is that LLMs present outputs fluently and confidently even when they are wrong or poorly grounded, which can encourage users to accept answers based on style rather than verification (Bodell, 2025). Going back to the findings of my thesis, this statement aligns well with it as it was found models correctly identified the true source of its answer only half of the time. When source attribution is unreliable yet delivered confidently, users are nudged toward deference rather than active evaluation, precisely the behavioural pathway through which cognitive offloading becomes a learning risk.

8.2 Reduced Learning

Cognitive offloading matters because learning is not the same thing as task completion. Learning typically involves effortful engagement, including struggling with ideas, making mistakes, and revising mental models, and this process is closely tied to motivation and long-term retention. A 2025 peer reviewed editorial examining the experiential effects of AI on learning argues that the satisfaction associated with learning is partly driven by an effort reward cycle, in which overcoming difficulty activates cognitive and motivational reinforcement mechanisms that support deeper understanding and sustained engagement (Hasan, 2025). A useful analogy can be drawn from game environments, where cheating may produce rapid outcomes but does not contribute to the learning process and replication without cheats would be impossible.

When AI tools deliver immediate solutions, such as generating essays or solving complex problems within seconds, they can bypass this effort reward cycle by shifting the learner from an active problem solver to a passive consumer. Although the resulting output may appear polished or correct, the depth of understanding is often reduced because the learner has not engaged with the reasoning process required to arrive at the solution independently (Hasan, 2025). The same analysis highlights the risk of an illusion of mastery, where learners feel

competent because tasks are completed, despite lacking the conceptual foundations needed to apply knowledge in new or unfamiliar contexts.

8.3 Educational Considerations

The impact of generative AI on learning environments depends less on the technology itself and more on how it is integrated into educational practice. In the Australian context, recent analysis highlights that AI tools can provide meaningful support to educators by reducing administrative workload, assisting with lesson planning, and enabling more personalised learning materials for students with diverse needs (Galovic, 2024). These applications are often framed as mechanisms for freeing educators to focus on activities that require human judgment, such as mentoring, feedback, and student engagement.

At the same time, both educators and policymakers have expressed concern that unstructured or permissive use of generative AI may undermine the development of foundational skills. Survey results from Australian teachers indicate that while many view AI as a useful support tool, a majority remain concerned about ethical implications, particularly the risk that students may use AI to complete assignments without engaging with the underlying material (Henebery, 2024). This concern is not limited to academic integrity, but extends to longer term learning outcomes, as reduced engagement with problem solving and reasoning can create gaps that compound over time.

These concerns suggest that the core challenge is not whether AI should be present in educational settings, but how its use is framed and assessed. We must remember that AI is here to stay, and the best we can do is moderate its usage. When AI is positioned as a substitute for student effort, it risks reinforcing cognitive offloading and shallow learning. When it is used as a scaffold, such as providing feedback, identifying areas of difficulty, or offering alternative explanations, it has the potential to support learning while preserving active cognitive engagement (Galovic, 2024).

Importantly, regulatory frameworks and ethical guidelines alone are insufficient to address these issues. While national and international policies emphasise principles such as transparency, fairness, and accountability, they do not directly address the experiential dimensions of learning. Preserving meaningful learning in the presence of generative AI requires deliberate teaching choices, including assessment designs that reward reasoning, explanation, and application rather than output alone. Without such structural changes, the convenience offered by AI risks displacing the very processes through which learning, skill development, and intellectual confidence are formed.

9. Recommendations

Generative AI is now embedded too deeply into modern life to meaningfully opt out. I grew up alongside tools like ChatGPT during my time at university and continued using them as I moved into professional work. Like many others, I have relied on generative AI extensively. It is fast, capable, and genuinely impressive. It helps explain complex ideas, draft content, and reduce friction in everyday tasks. However, the more I used it, the clearer it became that

these benefits come with hidden costs that are easy to ignore when the technology feels convenient and harmless.

Because of this, ethical responsibility cannot rest solely with companies or regulators. It must also extend to individuals who use these systems every day. The first recommendation is moderation. Not every task requires a generative AI response. Before submitting a query, individuals can pause and ask whether the task genuinely benefits from AI assistance or whether it replaces learning, critical thinking, or creativity that could reasonably be done independently. While the environmental cost of a single query is negligible in isolation, repeated and unnecessary use at scale contributes to cumulative harm. Conscious restraint is therefore one of the few forms of agency that users still retain.

That cumulative harm is not abstract. The data centres that power generative AI systems require vast amounts of energy and water, and in some cases have already been associated with environmental pollution and health risks borne by nearby communities. At the same time, recent misuse of generative AI, such as the rapid production of nonconsensual sexualised imagery, demonstrates that these systems can cause direct harm when safeguards fail. Being moderate in use is therefore not only an environmental consideration, but also an ethical one. Each interaction participates in a system whose impacts extend beyond the individual user.

Second, generative AI should be treated as an assistive tool rather than an authority. Outputs should be questioned, sources verified, and conclusions tested, particularly in educational, professional, or high-stakes contexts. Blind trust in fluent AI responses increases the risk of misinformation, shallow understanding, and misplaced confidence. Developing habits of verification is not about rejecting the technology but about preserving human responsibility for judgment and accountability.

Third, users should be cautious about what they share. Generative AI systems are not private spaces, even when they feel conversational or supportive. Avoiding the disclosure of sensitive personal, medical, legal, or emotional information is a practical ethical safeguard that individuals can apply immediately, regardless of platform promises or settings.

Fourth, respect for creative work must be part of responsible use. Many generative AI systems are trained on vast amounts of artistic and written material, often without the explicit consent of creators. While individuals may not control how these models are trained, they do control how the outputs are used. Choosing not to replace artists, writers, or designers with AI generated substitutes where human work is available is one-way users can acknowledge the value of creative labour rather than treating it as an unlimited resource.

Finally, individuals should actively educate themselves about how generative AI systems work and what their hidden costs are. Understanding the environmental footprint, data practices, and governance failures discussed in this paper changes how the technology is perceived and used. The goal is not to reject generative AI, but to engage with it deliberately, critically, and responsibly.

Taken together, these actions may seem insignificant at the individual level. However, ethical change in technological systems rarely begins with regulation alone. It begins with informed

users who understand the consequences of convenience and choose, when possible, not to optimise solely for speed, ease, or automation.

10. Conclusion

Generative AI is often presented as an inevitable technological progression: faster, smarter, and increasingly embedded into daily life. This paper has argued that while its capabilities are impressive, its ethical costs remain largely hidden from view. These costs span multiple dimensions, including unreliable sourcing, environmental degradation, artistic exploitation, privacy erosion, guardrail failure, and diminished learning. None of these harms arise from a single feature or isolated failure. Instead, they emerge from a broader pattern of rapid deployment in which speed, scale, and visibility are prioritised over precaution, transparency, and accountability. Within this pattern sits the myth of consent: users technically agree to the use of these systems yet do so without full knowledge of their downstream consequences, and often only discover the true costs after opting out has become difficult or impossible.

A central theme throughout this analysis is asymmetry. The benefits of generative AI are global, immediate, and highly visible. The harms are cumulative, unevenly distributed, and often borne by individuals and communities with limited power or choice. Creators see their work absorbed without meaningful consent, users surrender sensitive data under assumptions of privacy, students lose opportunities for deep learning, and marginalised communities absorb the environmental and health costs of infrastructure expansion. In each case, consent is implied rather than genuinely informed, reinforcing the gap between those who benefit from generative AI and those who bear its risks. As with earlier technological shifts, those who benefit most are rarely those who bear the greatest risk.

This paper does not argue that generative AI should be abandoned, nor does it seek to persuade readers to reject its use. Its aim is awareness. Once the hidden mechanisms and consequences of generative AI are made visible, continued use becomes a conscious decision rather than an automatic one. Ethical engagement with technology begins not with prohibition, but with understanding, restraint, and responsibility.

By synthesising technical architecture, case studies, and lived experience, this paper reframes generative AI ethics away from isolated failures or misuse and toward a systemic imbalance between visibility and consequence. The contribution is not the identification of new harms, but the demonstration that these harms are connected by a shared logic of obscured responsibility and assumed consent. Recognising this pattern is a necessary step toward more informed use, meaningful accountability, and ethical engagement before generative AI becomes fully unavoidable.

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